

Benchmark results — Persistences, BLEND, AR-OLS, and ELM cost-function variants

The runs use $N_{\text{hidden}} = 500$ and $N_{\text{draws}} = 100$, $\approx 35\,000$ samples (2 years, August 2020 – August 2022), without dimensionality reduction.

The ELM and AR-OLS inputs are the standardized window of the $LB = 48$ lags, augmented by 4 cyclical time features (sin/cos encoding of the hour of the day and of the day of the year) for the target time step.

Predictions are clipped to $[0, +\infty)$ because the models sometimes predicted negative values, which are physically impossible.

Most ELM variants select their hyperparameter(s) from a small grid by validation RMSE (20 % of the training set). The λ column reports the selected value (or the selected tuple for two-parameter variants, in the order stated in each variant’s script). **Bold** marks the best NICE ^{Σ} among the cyclic ELM variants for each table (ties bolded together).

Two BLEND variants. **BLEND**_{optim} (*optimized*): λ per phase learned by directly minimizing the MSE on the training set (closed-form OLS solution). **BLEND**_{corr}: λ derived from the phase-conditioned empirical correlation ($\lambda = (1 + \rho)/2$). Both variants use the same persistences P and P° .

All data vs. daytime only. Each horizon is reported twice: *All data* includes all 30 min test steps (nights, where $P \approx 0$, dominate in count) and Daytime only, which restricts the evaluation to steps where the solar elevation at Palaiseau (48.71°N, 2.22°E) exceeds 0°.

FH = 0.5 h

Table 1: All data – $FH = 0.5$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1169.0	0.490	~ 0	0.239	0.910	1.000	1.000	1.000	1.000
Persistence P [°]	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.793	1.966	1.938	1.899
BLEND _{corr}	0	–	–	–	1133.1	0.475	−0.002	0.233	0.916	0.974	0.969	0.967	0.970
BLEND _{optim}	48	–	–	–	1079.7	0.452	−0.014	0.216	0.923	0.901	0.924	0.931	0.918
AR-OLS	53	–	–	–	1012.4	0.424	+0.014	0.212	0.933	0.887	0.866	0.880	0.878
ELM (OLS)	27000	500	100	–	991.6	0.415	+0.021	0.212	0.935	0.884	0.848	0.861	0.865
ELM (Ridge)	27000	500	100	25	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (Rob. Risk)	27000	500	100	5	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (Tikhonov)	27000	500	100	25	987.9	0.414	+0.024	0.211	0.936	0.881	0.845	0.860	0.862
ELM (Box-Cox)	27000	500	100	(25,1)	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (MAE)	27000	500	100	(10,10)	983.7	0.412	+0.020	0.205	0.936	0.859	0.841	0.862	0.854
ELM (Log-MSE)	27000	500	100	(25,100)	1234.2	0.517	−0.043	0.238	0.900	0.993	1.056	1.073	1.041
ELM (Huber)	27000	500	100	25	983.3	0.412	+0.019	0.201	0.936	0.841	0.841	0.868	0.850
ELM (L3)	27000	500	100	(10,500)	1161.9	0.487	+0.050	0.288	0.911	1.202	0.994	0.936	1.044
ELM (Corr)	27000	500	100	(25,0.5)	997.6	0.418	+0.022	0.201	0.935	0.841	0.853	0.892	0.862
ELM (Elastic Net)	27000	500	100	(1,25)	986.8	0.413	+0.022	0.210	0.936	0.879	0.844	0.859	0.861
ELM (M-Est.)	27000	500	100	25	985.0	0.413	+0.019	0.200	0.936	0.835	0.843	0.871	0.850
ELM (Lp)	27000	500	100	(25,1.5)	982.9	0.412	+0.021	0.205	0.937	0.855	0.841	0.863	0.853
ELM (GLM)	27000	500	100	10	1115.3	0.467	+0.085	0.274	0.918	1.144	0.954	0.940	1.013
ELM _{no cyc} (OLS)	25000	500	100	–	1000.2	0.419	+0.020	0.211	0.934	0.881	0.856	0.873	0.870
ELM _{no cyc} (Ridge)	25000	500	100	25	992.6	0.416	+0.016	0.208	0.935	0.869	0.849	0.866	0.861

Table 2: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 0.5$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1650.0	0.347	~ 0	0.239	0.857	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.795	1.966	1.938	1.900
BLEND _{corr}	0	–	–	–	1599.3	0.336	–0.002	0.233	0.866	0.974	0.969	0.967	0.970
BLEND _{optim}	48	–	–	–	1524.0	0.320	–0.014	0.216	0.878	0.902	0.924	0.931	0.919
AR-OLS	53	–	–	–	1423.3	0.299	+0.001	0.200	0.894	0.836	0.863	0.880	0.859
ELM (OLS)	27000	500	100	–	1382.8	0.291	–0.001	0.189	0.900	0.793	0.838	0.860	0.830
ELM (Ridge)	27000	500	100	25	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (Rob. Risk)	27000	500	100	5	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (Tikhonov)	27000	500	100	25	1379.4	0.290	+0.002	0.189	0.900	0.791	0.836	0.859	0.829
ELM (Box-Cox)	27000	500	100	(25,1)	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (MAE)	27000	500	100	(10,10)	1376.6	0.289	+0.001	0.186	0.901	0.780	0.834	0.861	0.825
ELM (Log-MSE)	27000	500	100	(25,100)	1741.8	0.366	–0.046	0.235	0.841	0.983	1.056	1.073	1.037
ELM (Huber)	27000	500	100	25	1377.6	0.290	+0.002	0.184	0.900	0.768	0.835	0.868	0.823
ELM (L3)	27000	500	100	(10,500)	1576.5	0.331	+0.001	0.238	0.870	0.995	0.955	0.927	0.959
ELM (Corr)	27000	500	100	(25,0.5)	1394.6	0.293	+0.002	0.181	0.898	0.759	0.845	0.891	0.832
ELM (Elastic Net)	27000	500	100	(1,25)	1379.4	0.290	+0.001	0.190	0.900	0.793	0.836	0.858	0.829
ELM (M-Est.)	27000	500	100	25	1381.0	0.290	+0.002	0.183	0.900	0.766	0.837	0.871	0.825
ELM (Lp)	27000	500	100	(25,1.5)	1375.6	0.289	+0.002	0.186	0.901	0.777	0.834	0.862	0.824
ELM (GLM)	27000	500	100	10	1532.9	0.322	+0.035	0.224	0.877	0.936	0.929	0.936	0.934
ELM _{no cyc} (OLS)	25000	500	100	–	1397.1	0.294	–0.001	0.190	0.898	0.796	0.847	0.872	0.838
ELM _{no cyc} (Ridge)	25000	500	100	25	1390.8	0.292	–0.002	0.190	0.899	0.797	0.843	0.866	0.835

FH = 1 hTable 3: All data – $FH = 1$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	1699.5	0.712	~ 0	0.379	0.810	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.3	0.963	~ 0	0.429	0.653	1.133	1.352	1.429	1.305
BLEND _{corr}	0	–	–	–	1603.5	0.672	–0.004	0.357	0.831	0.941	0.944	0.945	0.943
BLEND _{optim}	48	–	–	–	1379.6	0.578	–0.031	0.283	0.875	0.746	0.812	0.848	0.802
AR-OLS	53	–	–	–	1277.0	0.535	+0.022	0.285	0.893	0.753	0.751	0.785	0.763
ELM (OLS)	27000	500	100	–	1207.0	0.506	+0.024	0.269	0.904	0.709	0.710	0.751	0.724
ELM (Ridge)	27000	500	100	25	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (Rob. Risk)	27000	500	100	5	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (Tikhonov)	27000	500	100	25	1203.0	0.504	+0.021	0.267	0.905	0.703	0.708	0.749	0.720
ELM (Box-Cox)	27000	500	100	(10,0.75)	1197.6	0.502	–0.019	0.246	0.906	0.648	0.705	0.756	0.703
ELM (MAE)	27000	500	100	(25,10)	1200.9	0.503	+0.015	0.259	0.905	0.683	0.707	0.753	0.714
ELM (Log-MSE)	27000	500	100	(25,100)	1405.4	0.589	–0.073	0.275	0.870	0.726	0.827	0.888	0.814
ELM (Huber)	27000	500	100	25	1203.1	0.504	+0.016	0.255	0.905	0.672	0.708	0.760	0.713
ELM (L3)	27000	500	100	(10,500)	1380.7	0.578	+0.053	0.356	0.875	0.939	0.812	0.796	0.849
ELM (Corr)	27000	500	100	(25,0.5)	1213.1	0.508	+0.027	0.262	0.903	0.692	0.714	0.771	0.726
ELM (Elastic Net)	27000	500	100	(10,25)	1202.0	0.503	+0.019	0.266	0.905	0.701	0.707	0.748	0.719
ELM (M-Est.)	27000	500	100	10	1214.2	0.509	+0.015	0.253	0.903	0.669	0.714	0.770	0.718
ELM (Lp)	27000	500	100	(25,1.25)	1199.7	0.502	+0.016	0.258	0.905	0.682	0.706	0.753	0.714
ELM (GLM)	27000	500	100	10	1288.5	0.540	+0.062	0.303	0.891	0.798	0.758	0.797	0.784
ELM _{no cyc} (OLS)	25000	500	100	–	1223.8	0.513	+0.021	0.270	0.902	0.712	0.720	0.763	0.732
ELM _{no cyc} (Ridge)	25000	500	100	25	1218.1	0.510	+0.016	0.266	0.903	0.702	0.717	0.762	0.727

Table 4: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 1$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	2397.6	0.504	–0.003	0.376	0.698	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.1	0.682	~ 0	0.429	0.448	1.140	1.353	1.429	1.308
BLEND _{corr}	0	–	–	–	2262.4	0.476	–0.006	0.354	0.731	0.941	0.944	0.945	0.943
BLEND _{optim}	48	–	–	–	1947.2	0.409	–0.031	0.282	0.801	0.750	0.812	0.848	0.804
AR-OLS	53	–	–	–	1792.6	0.377	+0.004	0.267	0.831	0.709	0.748	0.784	0.747
ELM (OLS)	27000	500	100	–	1682.6	0.354	–0.004	0.241	0.851	0.639	0.702	0.750	0.697
ELM (Ridge)	27000	500	100	25	1681.2	0.353	–0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (Rob. Risk)	27000	500	100	5	1681.2	0.353	–0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (Tikhonov)	27000	500	100	25	1680.3	0.353	–0.005	0.241	0.852	0.639	0.701	0.748	0.696
ELM (Box-Cox)	27000	500	100	(10,0.75)	1687.5	0.355	–0.029	0.236	0.851	0.627	0.704	0.756	0.696
ELM (MAE)	27000	500	100	(25,10)	1683.2	0.354	–0.005	0.238	0.851	0.632	0.702	0.753	0.696
ELM (Log-MSE)	27000	500	100	(25,100)	1983.4	0.417	–0.075	0.273	0.794	0.724	0.827	0.888	0.813
ELM (Huber)	27000	500	100	25	1686.6	0.355	–0.005	0.234	0.851	0.622	0.703	0.760	0.695
ELM (L3)	27000	500	100	(10,500)	1877.3	0.395	–0.005	0.298	0.815	0.791	0.783	0.789	0.787
ELM (Corr)	27000	500	100	(25,0.5)	1692.9	0.356	~ 0	0.235	0.850	0.624	0.706	0.770	0.700
ELM (Elastic Net)	27000	500	100	(10,25)	1681.2	0.353	–0.005	0.241	0.852	0.641	0.701	0.747	0.697
ELM (M-Est.)	27000	500	100	10	1704.6	0.358	–0.003	0.235	0.848	0.625	0.711	0.770	0.702
ELM (Lp)	27000	500	100	(25,1.25)	1681.1	0.353	–0.005	0.237	0.852	0.630	0.701	0.753	0.695
ELM (GLM)	27000	500	100	10	1796.5	0.378	+0.023	0.264	0.831	0.702	0.749	0.796	0.749
ELM _{no cyc} (OLS)	25000	500	100	–	1709.8	0.360	–0.005	0.244	0.847	0.649	0.713	0.762	0.708
ELM _{no cyc} (Ridge)	25000	500	100	25	1707.6	0.359	–0.006	0.244	0.847	0.649	0.712	0.762	0.708

FH = 3 hTable 5: All data – $FH = 3$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	3594.0	1.505	–0.001	0.890	0.151	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.5	0.963	~ 0	0.429	0.653	0.483	0.640	0.727	0.616
BLEND _{corr}	0	–	–	–	3007.6	1.260	–0.002	0.728	0.406	0.818	0.837	0.846	0.834
BLEND _{optim}	48	–	–	–	1914.3	0.802	–0.072	0.383	0.759	0.430	0.533	0.597	0.520
AR-OLS	53	–	–	–	1800.2	0.754	+0.029	0.425	0.787	0.478	0.501	0.540	0.506
ELM (OLS)	27000	500	100	–	1665.3	0.697	+0.026	0.385	0.818	0.433	0.463	0.506	0.468
ELM (Ridge)	27000	500	100	25	1660.3	0.695	+0.017	0.382	0.819	0.429	0.462	0.505	0.465
ELM (Rob. Risk)	27000	500	100	5	1660.3	0.695	+0.017	0.382	0.819	0.429	0.462	0.505	0.465
ELM (Tikhonov)	27000	500	100	25	1660.6	0.695	+0.021	0.383	0.819	0.430	0.462	0.505	0.466
ELM (Box-Cox)	27000	500	100	(25,0.75)	1663.4	0.697	–0.043	0.357	0.818	0.401	0.463	0.513	0.459
ELM (MAE)	27000	500	100	(25,100)	1673.6	0.701	+0.005	0.368	0.816	0.413	0.466	0.517	0.465
ELM (Log-MSE)	27000	500	100	(25,100)	1887.9	0.791	–0.145	0.378	0.766	0.424	0.525	0.590	0.513
ELM (Huber)	27000	500	100	25	1679.3	0.703	+0.005	0.364	0.815	0.409	0.467	0.521	0.466
ELM (L3)	27000	500	100	(10,500)	1822.3	0.763	+0.081	0.483	0.782	0.542	0.507	0.518	0.522
ELM (Corr)	27000	500	100	(25,0.5)	1660.9	0.696	+0.032	0.384	0.819	0.432	0.462	0.506	0.467
ELM (Elastic Net)	27000	500	100	(1,25)	1660.3	0.695	+0.017	0.382	0.819	0.429	0.462	0.505	0.465
ELM (M-Est.)	27000	500	100	25	1699.6	0.712	–0.003	0.360	0.810	0.404	0.473	0.531	0.470
ELM (Lp)	27000	500	100	(25,1.25)	1670.6	0.700	+0.010	0.370	0.817	0.416	0.465	0.515	0.465
ELM (GLM)	27000	500	100	10	1737.0	0.727	+0.056	0.405	0.802	0.455	0.483	0.535	0.491
ELM _{no cyc} (OLS)	25000	500	100	–	1713.1	0.717	+0.020	0.392	0.807	0.441	0.477	0.523	0.480
ELM _{no cyc} (Ridge)	25000	500	100	25	1705.2	0.714	+0.012	0.388	0.809	0.436	0.474	0.521	0.477

Table 6: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 3\text{ h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	4919.5	1.034	−0.082	0.808	−0.270	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.4	0.682	~ 0	0.429	0.448	0.531	0.660	0.738	0.643
BLEND _{corr}	0	–	–	–	4147.1	0.872	−0.059	0.671	0.098	0.831	0.843	0.850	0.841
BLEND _{optim}	48	–	–	–	2701.8	0.568	−0.073	0.382	0.617	0.473	0.549	0.606	0.543
AR-OLS	53	–	–	–	2520.8	0.530	−0.001	0.395	0.667	0.489	0.512	0.548	0.516
ELM (OLS)	27000	500	100	–	2326.0	0.489	−0.010	0.349	0.716	0.433	0.473	0.513	0.473
ELM (Ridge)	27000	500	100	25	2325.3	0.489	−0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (Rob. Risk)	27000	500	100	5	2325.3	0.489	−0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (Tikhonov)	27000	500	100	25	2323.3	0.488	−0.012	0.350	0.717	0.433	0.472	0.512	0.472
ELM (Box-Cox)	27000	500	100	(25,0.75)	2343.8	0.493	−0.055	0.344	0.712	0.426	0.476	0.520	0.474
ELM (MAE)	27000	500	100	(25,100)	2350.0	0.494	−0.020	0.342	0.710	0.424	0.478	0.524	0.475
ELM (Log-MSE)	27000	500	100	(25,100)	2664.7	0.560	−0.149	0.374	0.627	0.463	0.542	0.598	0.534
ELM (Huber)	27000	500	100	25	2357.8	0.496	−0.020	0.338	0.708	0.419	0.479	0.529	0.476
ELM (L3)	27000	500	100	(10,500)	2487.8	0.523	+0.008	0.410	0.675	0.507	0.506	0.521	0.511
ELM (Corr)	27000	500	100	(25,0.5)	2316.0	0.487	−0.006	0.346	0.719	0.428	0.471	0.512	0.471
ELM (Elastic Net)	27000	500	100	(1,25)	2325.3	0.489	−0.014	0.351	0.716	0.434	0.473	0.512	0.473
ELM (M-Est.)	27000	500	100	25	2388.8	0.502	−0.026	0.337	0.701	0.417	0.486	0.539	0.481
ELM (Lp)	27000	500	100	(25,1.25)	2344.2	0.493	−0.017	0.343	0.712	0.425	0.477	0.522	0.475
ELM (GLM)	27000	500	100	10	2433.9	0.512	+0.014	0.363	0.689	0.449	0.495	0.543	0.496
ELM _{no cyc} (OLS)	25000	500	100	–	2396.0	0.504	−0.016	0.356	0.699	0.441	0.487	0.530	0.486
ELM _{no cyc} (Ridge)	25000	500	100	25	2390.1	0.502	−0.019	0.356	0.700	0.441	0.486	0.529	0.485

FH = 6 h

Table 7: All data – $FH = 6\text{ h}$

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	5509.8	2.307	~ 0	1.510	−0.994	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.7	0.963	~ 0	0.429	0.653	0.284	0.417	0.494	0.398
BLEND _{corr}	0	–	–	–	3817.0	1.598	−0.005	1.017	0.043	0.673	0.693	0.706	0.691
BLEND _{optim}	48	–	–	–	2164.7	0.906	−0.115	0.423	0.692	0.280	0.393	0.463	0.379
AR-OLS	53	–	–	–	1960.5	0.821	+0.017	0.458	0.748	0.303	0.356	0.402	0.354
ELM (OLS)	27000	500	100	–	1879.7	0.787	+0.014	0.436	0.768	0.289	0.341	0.385	0.338
ELM (Ridge)	27000	500	100	25	1859.9	0.779	+0.007	0.425	0.773	0.282	0.338	0.382	0.334
ELM (Rob. Risk)	27000	500	100	10	1863.4	0.780	+0.001	0.425	0.772	0.281	0.338	0.383	0.334
ELM (Tikhonov)	27000	500	100	25	1871.9	0.784	+0.010	0.432	0.770	0.286	0.340	0.384	0.336
ELM (Box-Cox)	27000	500	100	(25,0.75)	1870.5	0.783	−0.055	0.405	0.770	0.268	0.339	0.387	0.332
ELM (MAE)	27000	500	100	(25,100)	1883.5	0.789	−0.005	0.413	0.767	0.274	0.342	0.393	0.336
ELM (Log-MSE)	27000	500	100	(25,100)	2057.9	0.862	−0.179	0.423	0.722	0.280	0.374	0.428	0.360
ELM (Huber)	27000	500	100	25	1895.0	0.793	−0.005	0.410	0.764	0.272	0.344	0.397	0.338
ELM (L3)	27000	500	100	(10,500)	2027.2	0.849	+0.064	0.530	0.730	0.351	0.368	0.389	0.369
ELM (Corr)	27000	500	100	(25,0.5)	1886.4	0.790	+0.020	0.439	0.766	0.291	0.342	0.386	0.340
ELM (Elastic Net)	27000	500	100	(10,25)	1859.9	0.779	+0.007	0.425	0.773	0.282	0.338	0.382	0.334
ELM (M-Est.)	27000	500	100	25	1916.2	0.802	−0.011	0.406	0.759	0.269	0.348	0.405	0.340
ELM (Lp)	27000	500	100	(25,1.25)	1878.4	0.787	~ 0	0.416	0.768	0.275	0.341	0.391	0.336
ELM (GLM)	27000	500	100	10	1935.2	0.810	+0.060	0.458	0.754	0.303	0.351	0.398	0.351
ELM _{no cyc} (OLS)	25000	500	100	–	1930.9	0.809	+0.007	0.448	0.755	0.297	0.350	0.398	0.348
ELM _{no cyc} (Ridge)	25000	500	100	25	1909.3	0.799	−0.002	0.439	0.761	0.291	0.347	0.393	0.343

Table 8: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 6$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6756.5	1.420	−0.408	1.103	−1.395	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.389	0.480	0.539	0.469
BLEND _{corr}	0	–	–	–	4911.6	1.032	−0.235	0.788	−0.266	0.714	0.727	0.735	0.725
BLEND _{optim}	48	–	–	–	3055.4	0.642	−0.115	0.423	0.510	0.383	0.452	0.505	0.447
AR-OLS	53	–	–	–	2750.6	0.578	−0.014	0.427	0.603	0.387	0.407	0.438	0.411
ELM (OLS)	27000	500	100	–	2631.2	0.553	−0.022	0.400	0.637	0.362	0.389	0.419	0.390
ELM (Ridge)	27000	500	100	25	2610.1	0.549	−0.023	0.396	0.643	0.359	0.386	0.417	0.387
ELM (Rob. Risk)	27000	500	100	10	2618.0	0.550	−0.025	0.399	0.640	0.362	0.387	0.418	0.389
ELM (Tikhonov)	27000	500	100	25	2624.1	0.552	−0.023	0.398	0.639	0.361	0.388	0.418	0.389
ELM (Box-Cox)	27000	500	100	(25,0.75)	2636.5	0.554	−0.068	0.392	0.635	0.356	0.390	0.422	0.389
ELM (MAE)	27000	500	100	(25,100)	2648.4	0.557	−0.029	0.389	0.632	0.352	0.392	0.428	0.391
ELM (Log-MSE)	27000	500	100	(25,100)	2904.6	0.611	−0.182	0.419	0.557	0.380	0.430	0.467	0.426
ELM (Huber)	27000	500	100	25	2664.0	0.560	−0.030	0.385	0.628	0.349	0.394	0.433	0.392
ELM (L3)	27000	500	100	(10,500)	2782.0	0.585	−0.006	0.460	0.594	0.417	0.412	0.420	0.416
ELM (Corr)	27000	500	100	(25,0.5)	2635.9	0.554	−0.020	0.400	0.635	0.362	0.390	0.420	0.391
ELM (Elastic Net)	27000	500	100	(10,25)	2610.1	0.549	−0.023	0.396	0.643	0.359	0.386	0.417	0.387
ELM (M-Est.)	27000	500	100	25	2696.0	0.567	−0.034	0.383	0.619	0.347	0.399	0.441	0.396
ELM (Lp)	27000	500	100	(25,1.25)	2639.8	0.555	−0.026	0.390	0.634	0.353	0.391	0.426	0.390
ELM (GLM)	27000	500	100	10	2712.4	0.570	+0.014	0.413	0.614	0.374	0.401	0.433	0.403
ELM _{no cyc} (OLS)	25000	500	100	–	2704.2	0.568	−0.031	0.409	0.616	0.371	0.400	0.433	0.402
ELM _{no cyc} (Ridge)	25000	500	100	25	2679.8	0.563	−0.035	0.406	0.623	0.368	0.397	0.429	0.398

FH = 10 hTable 9: All data – $FH = 10$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6403.1	2.680	~ 0	1.927	−1.693	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	2298.9	0.962	~ 0	0.429	0.653	0.223	0.359	0.440	0.341
BLEND _{corr}	0	–	–	–	3731.3	1.562	−0.004	1.110	0.086	0.576	0.583	0.592	0.584
BLEND _{optim}	48	–	–	–	2224.0	0.931	−0.134	0.433	0.675	0.225	0.347	0.423	0.332
AR-OLS	53	–	–	–	1964.9	0.823	+0.014	0.458	0.746	0.238	0.307	0.359	0.301
ELM (OLS)	27000	500	100	–	1894.3	0.793	+0.016	0.439	0.764	0.228	0.296	0.345	0.289
ELM (Ridge)	27000	500	100	25	1887.3	0.790	+0.008	0.434	0.766	0.225	0.295	0.344	0.288
ELM (Rob. Risk)	27000	500	100	10	1888.8	0.791	−0.003	0.432	0.766	0.224	0.295	0.344	0.288
ELM (Tikhonov)	27000	500	100	25	1889.3	0.791	+0.011	0.436	0.766	0.226	0.295	0.344	0.288
ELM (Box-Cox)	27000	500	100	(25,0.75)	1899.9	0.795	−0.061	0.412	0.763	0.214	0.297	0.349	0.286
ELM (MAE)	27000	500	100	(25,100)	1903.5	0.797	−0.011	0.417	0.762	0.217	0.297	0.352	0.289
ELM (Log-MSE)	27000	500	100	(25,100)	2087.2	0.874	−0.189	0.433	0.714	0.225	0.326	0.383	0.311
ELM (Huber)	27000	500	100	25	1914.6	0.801	−0.011	0.414	0.759	0.215	0.299	0.356	0.290
ELM (L3)	27000	500	100	(10,500)	2078.3	0.870	+0.062	0.549	0.716	0.285	0.325	0.353	0.321
ELM (Corr)	27000	500	100	(25,0.5)	1900.7	0.796	+0.014	0.440	0.763	0.228	0.297	0.345	0.290
ELM (Elastic Net)	27000	500	100	(10,25)	1887.3	0.790	+0.008	0.434	0.766	0.225	0.295	0.344	0.288
ELM (M-Est.)	27000	500	100	25	1931.0	0.808	−0.017	0.410	0.755	0.213	0.302	0.361	0.292
ELM (Lp)	27000	500	100	(25,1.25)	1900.7	0.796	−0.006	0.420	0.763	0.218	0.297	0.351	0.289
ELM (GLM)	27000	500	100	10	1972.5	0.826	+0.051	0.458	0.744	0.238	0.308	0.362	0.302
ELM _{no cyc} (OLS)	25000	500	100	–	1948.3	0.816	+0.004	0.451	0.751	0.234	0.304	0.355	0.298
ELM _{no cyc} (Ridge)	25000	500	100	25	1940.2	0.812	−0.003	0.444	0.753	0.231	0.303	0.354	0.296

Table 10: Daytime only ($\theta_{\text{sol}} > 0^\circ$) – $FH = 10$ h

Method	N_θ	N_h	N_d	λ	RMSE (W)	nRMSE	nMBE	nMAE	R^2	NICE ¹	NICE ²	NICE ³	NICE ^{Σ}
Persistence P	0	–	–	–	6688.9	1.406	–0.852	1.074	–1.348	1.000	1.000	1.000	1.000
Persistence P ^o	0	–	–	–	3244.6	0.682	~ 0	0.429	0.448	0.399	0.485	0.540	0.475
BLEND _{corr}	0	–	–	–	4262.7	0.896	–0.441	0.672	0.047	0.626	0.637	0.649	0.637
BLEND _{optim}	48	–	–	–	3138.9	0.660	–0.135	0.433	0.483	0.403	0.469	0.519	0.464
AR-OLS	53	–	–	–	2758.3	0.580	–0.016	0.429	0.601	0.399	0.412	0.440	0.417
ELM (OLS)	27000	500	100	–	2653.2	0.558	–0.019	0.404	0.631	0.376	0.397	0.423	0.398
ELM (Ridge)	27000	500	100	25	2648.3	0.557	–0.022	0.404	0.632	0.376	0.396	0.422	0.398
ELM (Rob. Risk)	27000	500	100	10	2655.3	0.558	–0.027	0.407	0.630	0.379	0.397	0.422	0.399
ELM (Tikhonov)	27000	500	100	25	2649.2	0.557	–0.021	0.404	0.632	0.376	0.396	0.422	0.398
ELM (Box-Cox)	27000	500	100	(25,0.75)	2678.6	0.563	–0.072	0.401	0.624	0.373	0.400	0.428	0.401
ELM (MAE)	27000	500	100	(25,100)	2678.7	0.563	–0.033	0.396	0.624	0.368	0.400	0.432	0.400
ELM (Log-MSE)	27000	500	100	(25,100)	2945.6	0.619	–0.192	0.430	0.545	0.400	0.440	0.471	0.437
ELM (Huber)	27000	500	100	25	2694.0	0.566	–0.033	0.392	0.619	0.365	0.403	0.437	0.402
ELM (L3)	27000	500	100	(10,500)	2845.0	0.598	–0.013	0.474	0.575	0.442	0.425	0.429	0.432
ELM (Corr)	27000	500	100	(25,0.5)	2662.7	0.560	–0.020	0.406	0.628	0.378	0.398	0.424	0.400
ELM (Elastic Net)	27000	500	100	(10,25)	2648.3	0.557	–0.022	0.404	0.632	0.376	0.396	0.422	0.398
ELM (M-Est.)	27000	500	100	25	2718.6	0.571	–0.037	0.390	0.612	0.363	0.406	0.444	0.404
ELM (Lp)	27000	500	100	(25,1.25)	2673.5	0.562	–0.029	0.397	0.625	0.369	0.400	0.431	0.400
ELM (GLM)	27000	500	100	10	2769.9	0.582	+0.015	0.422	0.597	0.393	0.414	0.444	0.417
ELM _{no cyc} (OLS)	25000	500	100	–	2734.2	0.575	–0.029	0.417	0.608	0.388	0.409	0.436	0.411
ELM _{no cyc} (Ridge)	25000	500	100	25	2726.5	0.573	–0.032	0.416	0.610	0.387	0.408	0.435	0.410